Roll Number: _____

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Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science & Engineering

MID SEMESTER EXAMINATION (2024-25 Even)

B. E. - IV Year, VIII Sem (COE / CSE / CSBS) Course Code: UCS813 / UCT802

Mar 04, 2025, Tuesday, 11:00 – 01:00 PM Course Name: Social Network Analysis

Time: 2 Hours, M. Marks: 30

Name Of Faculty: Dr. Ashutosh Aggarwal, Dr. Nitin Saxena

Note: *Attempt **all** questions. Assume missing data, if any, suitably.*

Q.1 (a) Given the set of nodes V, with |V| = 6, in which each node i is labelled by a natural number between 1 and 6, i = 1, 2, 3, 4, 5, 6, consider the directed network G = (V, E) where each link from node j to node i indicates that j is a multiple of i. Draw the network and write down the adjacency matrix of the network. (2)

(b) It is known that T is a tree with 10 vertices of degree 10 and all other vertices are leaves. How many vertices does T have? (2) [CO1 - BL1]

(c) A delivery driver Mr. X delivers for a food chain company. He has room to carry four deliveries in his car. He picks up four meals and sets out on his final delivery route of the day. He leaves his friend at point *A* and wants to deliver the meals before heading directly home to point *H* after the final delivery. *Figure 1* shows the positions of the houses (B, C D & E) to which Mr. X will deliver and the travel time between them, in minutes, on a weighted graph (not to scale). Mr. X does not wish to travel along the same street twice. What is the shortest route he can take to deliver the four meals and continue to home? (2)

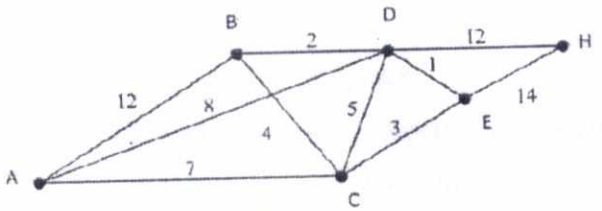


Figure 1

Q.2 (a) Consider the structure of the network shown in *figure 2* and answer the following questions: (4)

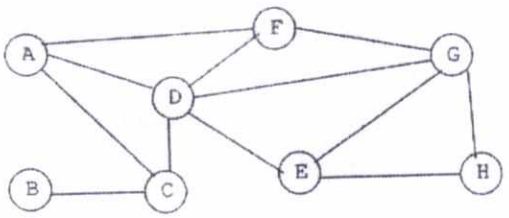


Figure 2

[CO2 - BL2]

i. Explain the concept of density and calculate the density of graph.
ii. Calculate networks' mean degree.
iii. Which vertices are the most far apart from each other? What is the longest path?

(b) Consider an undirected network of N nodes which is connected and has the largest diameter? Does this network have the small-world distance property? Why? (2)

Q.3 (a) In the social network depicted in *figure 3*, with each edge labeled as either a strong or weak tie, which nodes satisfy the Strong Triadic Closure Property and which do not? Provide an explanation for your answer. **(4)**

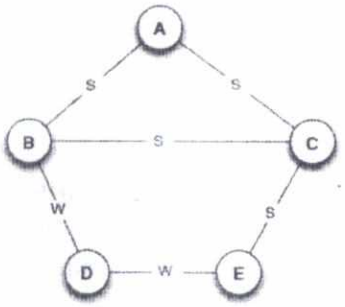
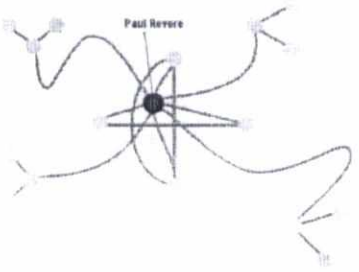


Figure 3

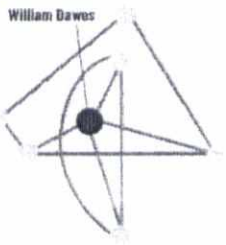
[CO1 – BL2]

(b) What is a scale-free network model? Why are those kinds of models important to study social networks? **(2)**

Q.4 (a) *Figures 4(a) & 4(b)* summarize network connections of Paul Revere and William Dawes respectively. Both rode from Boston in April 1775 spreading the news that the British army was moving to Lexington. **(4)**



(a)



(b)

Figure 4

[CO2 – BL2]

Who was more effective in spreading the message? Explain your answer.

(b) Calculate the Simple Matching Coefficient and the Jaccard coefficient similarity measures for the following two binary vectors: **(2)**

P1=(1, 0, 1, 0, 0, 0, 1, 0, 1, 0) P2=(0, 0, 1, 0, 1, 1, 1, 0, 0, 1)

Round your answer to two decimal places.

Q.5 What is page rank in context of world wide web? Why is page importance rating important? **(6)**

Figure 5 shows the link structure of four web pages. Calculate page rank using matrix multiplication method for the given graph. Perform computation for two iterations only.

[CO2 – BL3]

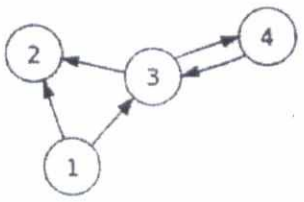


Figure 5

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